A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 04

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 5 points below. **1 Double hull:** a gap of up to 2 m between the hulls so damage to the outer hull in a collision / grounding does not release oil **1 Inert gas system:** cooled engine exhaust gas (no oxygen) fills the space above the oil so an explosive oil vapour / air mixture cannot form **1 Separate ballast tanks:** ballast water is not carried in oil tanks so no residual oil is discharged when ballast is pumped out **1 Twin engines / rudders:** if one fails the ship can still be steered / propelled and is not carried onto rocks by currents or wind **1 Crude oil washing:** sludges are mixed and removed with the cargo so tanks do not need washing with seawater that was then discharged – oily water is unloaded at the terminal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Offshore routes keep tankers further from the coast so if there is engine failure there is more time to deal with it before the ship is carried ashore **1** GPS gives accurate navigation – AIS lets ship positions be monitored and coordinated to avoid collisions and groundings **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** The tank stands on an impermeable base surrounded by a wall **1** The volume enclosed by the wall is greater than the tank volume so all the oil is contained if the tank splits / cannot overflow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through	2
01.4	1 mark for the point below. **1** Separates floating oil from drainage water so the oil is retained (for later treatment) and only water flows to the drains / river **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 259-260; AQA spec 3.4.3.2.7.	
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (4.8 - 1.4) / 4.8 x 100 **1** = 70.8% (accept 71%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** A tributary joins the river, more than doubling the flow (15 to 36 $m^3 s^{-1}$) **1** So the pollutant is diluted by the extra water (dispersal), not broken down **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** The pollutant is persistent so it is not degraded / broken down as it travels **1** Little extra water joins the river (flow 36 to 38) so little further dilution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	2
02.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Lake has a long residence time / water does not flow away so the pollutant accumulates rather than being carried away **1** Smaller volume of water / less dilution / less dispersal by currents **1** Persistent pollutant gradually becomes more concentrated (eg by evaporation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime • **Persistent** → does not break down / stays in the environment for a long time	2
02.5	**1 mark each** 1 mark for each of the 2 points below. **1 Marine discharge:** large volume of water and currents dilute and disperse the pollutant to acceptably low concentrations (control of emission location) **1 River:** small volume so concentrations stay high / drinking water abstraction / sensitive wildlife downstream **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abstraction** → water taken out of a river, lake or aquifer	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-241, 254-255; AQA spec 3.4.1, 3.4.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** They store chemical energy that is easy to keep until it is needed, and easy to convert into the heat that is usually wanted **1** They have a high energy density, so they reach the high temperatures needed to smelt metal ores and to raise high-pressure steam for turbines **1** High energy density also makes them efficient to transport, because a small mass or volume carries a lot of energy – 75 t of aviation fuel can carry a 400 t aircraft 5600 km **1** They were often found in very abundant local deposits, so industry could grow up on the coalfield that supplied it **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 75 t / 400 passengers = 0.1875 t per passenger **1** = 188 kg (accept 187 - 190 kg) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (kg) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
03.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Many deposits are too deep, too small or in places that are too difficult to reach for the technology available **1** A lot of natural gas is held in fine-grained impermeable shale, from which it cannot flow to a conventional well **1** Oil shale holds more than the world's total reserves of crude oil, but the hydrocarbons are solid and insoluble, so the rock has to be heated to melt the oil before it will flow **1** Reserves count only what can be extracted economically, so deposits that would cost more than the fuel is worth are excluded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through	3

Do not write outside the box

Question	Marking guidance	Marks
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Economic decisions are made on the cheapest option available at the time rather than on long-term supply, so the mines closed while coal was still in the ground **1** The abandoned mines have flooded and are unsafe, so the remaining coal cannot be reached even though North Sea gas will be exhausted within the next thirty or forty years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.5	**Any 5 of 7** 1 mark each for any 5 of the 7 points below, in any order, maximum 5. **1** Deep mining damages the surface through loss of habitat, the transport infrastructure it needs and subsidence above the worked seams **1** Opencast mining causes greater habitat damage because all the overburden above the coal has to be cleared **1** Spoil heaps from mining occupy land, can be unstable and are slow to revegetate **1** Oil spills from rigs pollute the sea, and oil-based drilling mud used to lubricate the drill pipes can pollute groundwater, rivers and the sea **1** Surplus gas on oil rigs is flared to reduce the risk of explosion, releasing sulfur dioxide, carbon dioxide and smoke **1** Pipeline construction damages habitat over a long, wide corridor, although careful route planning, soil replacement and replanting reduce this **1** Fracking to release shale gas uses large volumes of water, risks contaminating groundwater and can cause local earth tremors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Fracking** → pumping fluid into rock at high pressure to release gas or oil	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-190; AQA spec 3.3.4.1.	
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** The human ear is sensitive to a very wide range of sound pressures / powers (about 10^{12} range) **1** A logarithmic scale compresses this into a convenient range of numbers: each 10 dB increase represents a ten-fold increase in sound intensity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Difference = 30 dB = 3 x 10 dB **1** = 10^3 = 1000 times **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** 88 dB **1** Doubling the sound intensity increases the level by 3 dB (logarithmic scale) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	1 mark for the point below. **1** 0 dB is set as the quietest sound that can be heard (at 1000 Hz) - the threshold of human hearing – quieter sounds exist but cannot be heard **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **threshold** → the concentration below which no harm is detected	1
04.5	**1 mark each** 1 mark for each of the 3 points below. **1** 20 to 20 000 Hz **1** dB(A) weights sounds between about 1000 and 4000 Hz more heavily **1** Because the nerve cells / hair cells in the cochlea are more sensitive to these frequencies so they are heard more easily / cause more annoyance – range narrows with age or noise damage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-287; AQA spec 3.4.3.2.13.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (68 000 - 7 000) / 68 000 x 100 **1** = 89.7% (accept 90%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued **1** Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.3	**1 mark each** 1 mark for each of the 4 points below. **1** Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress – pollution by pesticides, fertilisers and industrial waste concentrated in the lake **1** Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) **1** Loss of wetland habitats and wildlife – local climate more extreme (loss of the moderating effect of the water body) **1** Loss of livelihoods – conflict between the countries sharing the rivers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive • **salinisation** → salt building up in the soil / soil becoming salty **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4

Do not write outside the box

Question	Marking guidance	Marks
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Upstream abstraction for irrigation reduces the flow available downstream **1** Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream – growing populations increase demand **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abstraction** → water taken out of a river, lake or aquifer • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
06.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (120 + 95) - (2 + 45) = 168 t ha^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO2 **1** Harvest removes biomass so less dead organic matter is returned – pasture roots and litter constantly add organic matter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Low / zero tillage **1** Adding manure, compost or mulch **1** Growing cover crops / leaving crop residues **1** Converting to long-term grass or tree crops in rotation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	
07.1	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1 Topography:** a narrow exit from a large deep basin so a small dam holds a large volume **1 Geology:** impermeable rock beneath (no seepage), strong enough to support the dam, no faults / seismic activity **1 Catchment area:** large enough with regular, reliable rainfall / river inflow – climate not too hot (evaporation losses) **1 Catchment land use:** low pollution risk (industry, pesticides) – low soil erosion (sedimentation reduces capacity) **1 Infrastructure / proximity to demand:** access for construction, cost of transporting water **1 Existing land use:** value of what is lost (urban areas, conservation sites, farmland) - loss:benefit analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
07.2	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2 Advantage:** fewer land-use conflicts as no land is flooded – large volume near coastal cities **up to 2 Disadvantages:** destroys / changes important intertidal habitats (mudflats for wading birds) – obstacle to shipping – collects pollution from the whole river catchment – sedimentation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Very energy-intensive and expensive (reverse osmosis / distillation) **1** Only worthwhile where seawater is available and freshwater supplies are inadequate (eg Malta, Middle East) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
07.4	1 mark for the point below. **1** Surplus water in wet periods (when infiltration capacity is exceeded and runoff would go to the sea) is pumped underground or diverted into lagoons to infiltrate, so abstraction in dry periods is balanced by recharge **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs • **infiltration** → water soaking into the soil surface **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-124; AQA spec 3.2.2.5.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Aerobic decomposition produces little methane and returns nutrients to the soil	1
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** It may contain concentrated heavy metals / toxic residues **1** It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
08.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Impermeable polymer / clay liner to stop leachate escaping **1** Collection and treatment of leachate **1** Collection of methane and use as fuel **1** Daily covering of waste / final capping and restoration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Impermeable** → does not let water through • **leachate** → polluted liquid draining out of waste **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** The energy used to extract raw materials, manufacture the components and assemble the vehicle (for a typical car about 1.5 years of its fuel use) **1** Reduced by using recycled materials (eg recycled aluminium needs 1/20 of the energy of new) **1** Lighter designs using less material – longer vehicle life spreading the embodied energy **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	3
09.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Use of recyclable materials wherever possible **1** Easy identification of components and their composition (code stamps) **1** Easy dismantling and separation of components / materials **1** Reusable components for new vehicles **1** Compostable materials for parts that cannot be recycled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** At low speed a large proportion of the energy just runs the engine rather than moving the car **1** At high speed energy is wasted overcoming air resistance (which rises with the square of speed) – optimum for many cars about 56 mph **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.4	1 mark for the point below. **1** Automatic stop / start systems cutting the engine when stationary – smooth driving in higher gears avoiding sudden braking / acceleration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 227-228; AQA spec 3.3.4.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Renewables are intermittent (solar at night, calm days, slack tides) and some unpredictable – demand fluctuates daily, weekly, seasonally and short-term – supply and demand rarely match • Without storage, surplus is wasted and shortages must be met by fossil-fuel peaking plants – large-scale electricity storage not yet possible directly so energy must be converted • **Pumped-storage HEP:** proven, rapid response, peak shaving, but limited sites • **Batteries:** lithium-ion for grid and vehicles – factors affecting viability – V2G • **Hydrogen:** electrolysis, storage methods, fuel cells, hydrogen economy, P2G into the gas grid – high energy density enables vehicle use • Compressed air with heat recovery – thermal storage (molten salt for CSP, high-volume stores, thermal mass) – flywheels – supercapacitors • Grid management / ICT and diversity of resources (wind + solar + tidal + biofuel) reduce but do not remove the need for storage – energy conservation reduces peaks • Evaluation of efficiency losses, costs, resource limits (lithium), and which technologies suit short-term vs inter-seasonal storage – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous • **HEP** → electricity generated from falling water • **energy conservation** → using less energy, eg by insulation or behaviour change **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	
	Model answer - Level 3 (9 marks)	
	Renewable electricity is intermittent - solar stops at night, wind on calm days, tidal power at slack water - and some of it is unpredictable, while demand fluctuates by the minute, day, week and season, so supply and demand rarely coincide. Without storage, surplus renewable output is wasted and shortfalls must be met by fossil-fuel peaking plants, which defeats the purpose. Electricity itself cannot be stored at scale, so the energy must be converted into another form and back.	
	Pumped-storage HEP is proven and responds within seconds for peak shaving, but suitable sites are few. Lithium-ion batteries now work at grid scale and in vehicles, with vehicle-to-grid schemes turning car batteries into distributed storage, though lithium supplies and lifetime limit them. Hydrogen from electrolysis can be stored, used in fuel cells or injected into the gas grid (power-to-gas), and its high energy density suits vehicles, although each conversion loses energy. Compressed air with heat recovery, molten-salt thermal stores for concentrating solar plants, thermal mass in buildings, flywheels and supercapacitors each fit particular timescales.	
	Storage is not the only answer: a diverse mix of wind, solar, tidal and biofuel smooths supply, smart-grid management shifts demand, and conservation reduces peaks. But these reduce rather than remove the mismatch. Short-term storage can be batteries and pumped hydro; the harder problem is seasonal storage, where only hydrogen or fuels made from surplus electricity look plausible. Without both, renewables can supplement but not dominate a country's supply.	
	Weaker answer - Level 2 (5 marks)	
	Renewable energy is intermittent. Solar panels do not work at night and wind turbines do not work when it is not windy, but people still need electricity then. Storage means the energy can be saved for when it is needed. Pumped storage pumps water uphill when there is spare electricity and lets it flow back through turbines at peak times. Batteries can also store electricity. Hydrogen can be made from electricity and used in fuel cells. So storage is needed for renewables to provide most electricity.	
	What makes the difference	
	• Level 2 explains the mismatch and describes three storage methods correctly, but does not evaluate them (sites, lithium, conversion losses, timescales).	
	• Level 3 distinguishes short-term from seasonal storage - the key insight for 'most of a country's electricity'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 also considers the alternatives to storage (diversity, grid management, conservation) and explains why they are insufficient alone.	
10.2	1 mark for the point below. **1** Storing surplus energy at times of low demand / high supply to meet later peaks in demand, so peak generating capacity can be reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 217-224; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Scale of the problem:** how energy is used (transport, buildings, industry, water) – links between affluence and per capita use • **Transport:** bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • **Buildings:** orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • **Industry:** heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • **Limits:** rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples – comparison with the cost of new supply	25
	– conclusion that conservation is the cheapest first step but cannot alone meet demand growth *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • **Fossil fuels:** choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports – new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • **Nuclear:** high energy density reduces transport – waste properties (half-life) determine storage – thorium and fusion prospects • **Renewables:** matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting) – intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity – low-impact designs (helical turbines, VAWT, tidal lagoons) • **Energy conservation reduces demand:** transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use	25
	– life-cycle / embodied energy • **Evaluation:** no resource is impact-free – knowledge allows the best mix and mitigation – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

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